

1)

	D	H	Q	RPM
Prototype	1	200	4.1	1200
Model	0.2			

Using the flow coefficient:

$$C_Q = \frac{Q}{\omega D^3}$$

$$\left(\frac{Q}{\omega D^3} \right)_m = \left(\frac{Q}{\omega D^3} \right)_p$$

$$\frac{Q_m}{1 \times 0.2^3} = \frac{4.1}{1 \times 1^3}$$

$$Q_m = 0.0328 \text{ m}^3 \text{ s}^{-1}$$

Using the head coefficient:

$$C_H = \frac{gH}{\omega^2 D^5}$$

$$\left(\frac{gH}{\omega^2 D^5} \right)_m = \left(\frac{gH}{\omega^2 D^5} \right)_p$$

$$\frac{H_m}{0.2^2 \times 1^2} = \frac{200}{1^2 \times 1^2}$$

$$H_m = 8 \text{ m}$$

2)	D	ω	Q	H
Model	0.25	1	?	?
Prototype	0.30	1	0.2	18.3

Using the flow coefficient:

$$C_Q = \frac{Q}{\omega D^3}$$

$$\left(\frac{Q}{\omega D^3} \right)_m = \left(\frac{Q}{\omega D^3} \right)_p$$

$$\frac{Q_m}{0.25^3} = \frac{0.2}{0.3^3}$$

$$Q_m = \frac{25}{216} \approx 0.116 \text{ m}^3 \text{ s}^{-1}$$

Using the head coefficient:

$$C_H = \frac{gH}{\omega^2 D^2}$$

$$\left(\frac{gH}{\omega^2 D^2} \right)_m = \left(\frac{gH}{\omega^2 D^2} \right)_p$$

$$\frac{H_m}{0.25^2} = \frac{18.3}{0.30^2}$$

$$H_m = \frac{305}{24} \approx 12.708 \text{ m}$$

2) Using the power coefficient:

$$C_p = \frac{W}{\rho \omega^3 D^5}$$

$$\left(\frac{W}{\rho \omega^3 D^5} \right)_m = \left(\frac{W}{\rho \omega^3 D^5} \right)_p$$

$$\frac{W_m}{0.25^5} = \frac{44.7}{0.30^5}$$

$$W_m = 17.96392747$$
$$\approx 17.963 \text{ kW}$$

3) Energy equation from bottom tank to the top tank:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 + h_p - h_L = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$h_L = h_p - z_2$$

$$= 52 - 1010Q^2 - (30 + 3)$$

$$= 19 - 1010Q^2$$

$$\frac{V^2}{2g} = \frac{Q^2}{2gA^2} = \frac{Q^2}{2g\left(\frac{\pi D^2}{4}\right)^2}$$

$$= \frac{8Q^2}{g\pi^2 D^4}$$

$$h_L = f \frac{L}{D} \frac{8Q^2}{g\pi^2 D^4} + K_L \frac{8Q^2}{g\pi^2 D^4}$$

$$= \frac{8Q^2}{g\pi^2 D^4} \left(f \frac{L}{D} + K \right)$$

$$= \frac{8Q^2}{9.81\pi^2 (0.1)^4} \left(0.02 \left(\frac{30}{0.1} \right) + K \right)$$

$$= \frac{80000}{9.81\pi^2} Q^2 (6 + K)$$

When $K=1$,

$$\frac{80000}{9.81\pi^2} Q^2 (6+1) = 19 - 1010Q^2$$

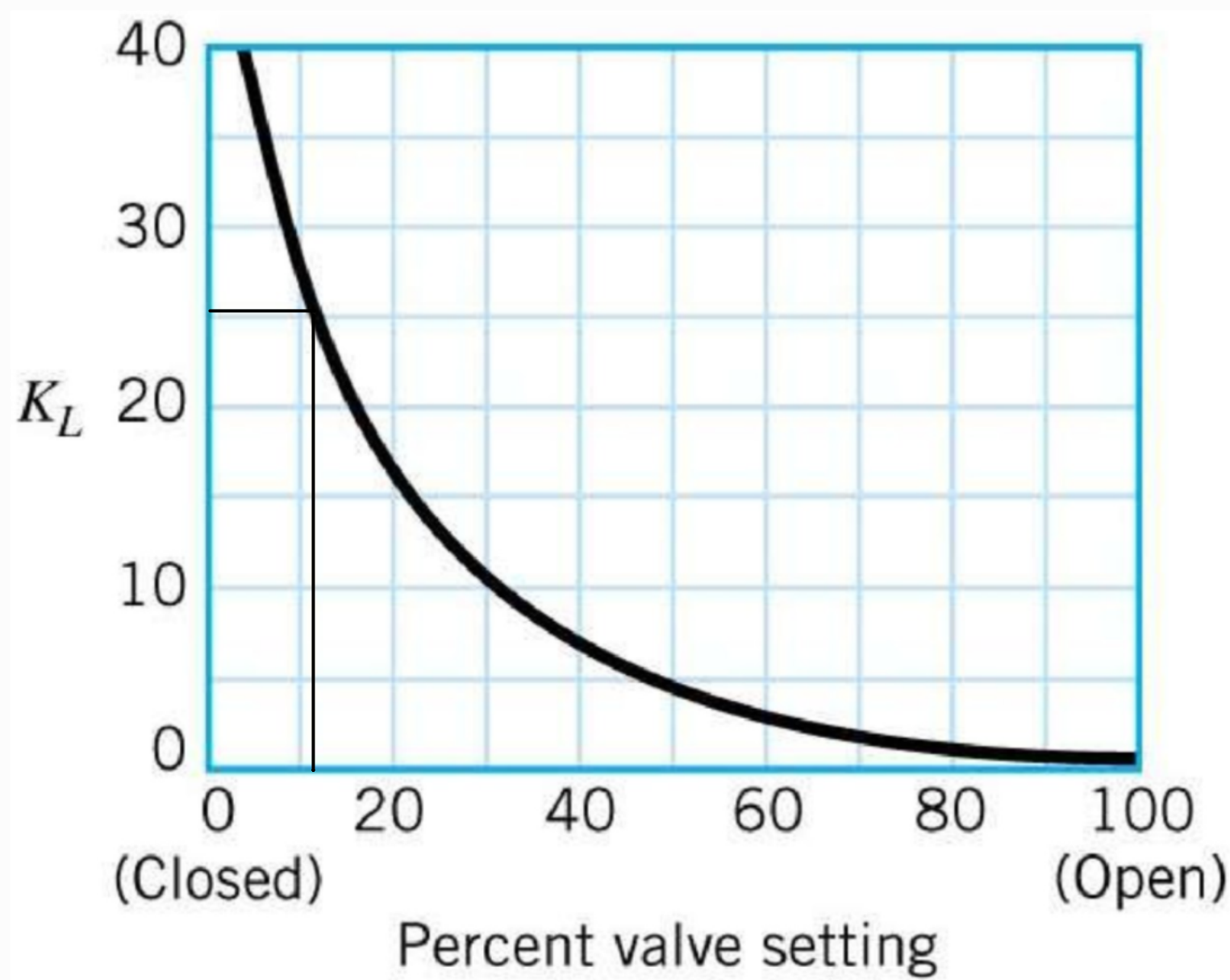
$$Q = 0.05288321678$$

$$\approx 0.052883 \text{ m}^3 \text{ s}^{-1}$$

$$\begin{aligned}
 3) \text{ When } Q &= 0.052883 \times \frac{1}{2} \\
 &= 0.02644160839 \\
 &\approx 0.02644 \text{ m}^3 \text{ s}^{-1}
 \end{aligned}$$

$$\frac{80000}{9.81 \text{ m}^2} (0.02644)^2 (6+K) = 19 - 1010(0.02644)^2$$

$$\begin{aligned}
 K &= 25.66708853 \\
 &\approx 25.667
 \end{aligned}$$



From the graph, the valve opening is 10%.

$$4) \rho = 850 \text{ kg m}^{-3}$$

$$\frac{V^2}{2g} = \frac{Q^2}{2gA^2} = \frac{Q^2}{2g\left(\pi\frac{D^2}{4}\right)} = \frac{8Q^2}{g\pi^2 D^4}$$

$$h_L = f \frac{L}{D} \frac{V^2}{2g} + \sum K \frac{V^2}{2g}$$

$$= \frac{V^2}{2g} \left(f \frac{L}{D} + \sum K \right)$$

$$= \frac{8Q^2}{g\pi^2 D^4} \left(f \frac{L}{D} + \sum K \right)$$

$$= \frac{8}{9.81\pi^2 (0.08)^4} \left(0.012 \left(\frac{80}{0.08} \right) + 10 \right) Q^2$$

$$= 44379.65963 Q^2$$

$$\approx 44380 Q^2$$

Energy equation from A to B:

$$h_s + \frac{P_A}{\rho g} + \frac{V_A^2}{2g} + z_A - h_L = \frac{P_B}{\rho g} + \frac{V_B^2}{2g} + z_B$$

$$h_s = z_B - z_A + h_L$$

$$= 10 - 15 + 44380 Q^2$$

$$= -5 + 44380 Q^2$$

$\therefore$ The system characteristic is:

$$h_s = -5 + 44380 Q^2$$

5) Let $x = Q^2$, and obtaining the linear regression for the pump curve in the form:

$$h_p = a + bx$$

$$h_p = 50.85447835 - 11125.54113x \\ \approx 50.85 - 11126 Q^2$$

At the operating point:

$$h_s = h_p$$

$$-5 + 44380 Q^2 = 50.85 - 11126 Q^2$$

$$Q = 0.03503486324$$

$$\approx 0.035 \text{ m}^3 \text{ s}^{-1}$$

$$h_p = -5 + 44380 (0.035)^2 = 49.4734423 \\ \approx 49.47 \text{ m}$$

Obtaining the quadratic regression of η against Q :

$$\eta = aQ^2 + bQ + c$$

$$= -39404.7614Q^2 + 3524.285714Q \\ - 0.202380952$$

When $Q = 0.035$:

$$\eta = -39405 (0.035)^2 + 3524.29 (0.035) \\ - 0.2024$$

$$= 74.9034419 \approx 74.9\%$$

$$\begin{aligned}
 5) \quad P &= \frac{\rho g Q h_p}{\eta} \\
 &= \frac{850 \times 9.81 (0.035) (49.47)}{74.9\%} \\
 &= 19295.61909 \text{ W} \\
 &\approx 19.30 \text{ kW}
 \end{aligned}$$

When an additional pump is added in series:

$$\begin{aligned}
 h_s &= 2h_p \\
 -5 + 44380 Q^2 &= 2(50.85 - 11126 Q^2) \\
 Q &= 0.04783726142 \\
 &\approx 0.0478 \text{ m}^3 \text{ s}^{-1}
 \end{aligned}$$